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Reg No. Name:

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

First Semester M. Tech Degree Examination January 2025

Branch: Information Technology **Specialization: Artificial Intelligence and Data Science**

Course Code: 221ECS004

Course Name: COMPUTATIONAL INTELLIGENCE

Max Marks: 60 **Duration: 2.5 Hours**

Q. No.	Question	Marks
	PART A (Answer all questions 5* 5 = 25 marks)	
1	Consider the following (discrete) fuzzy sets: A = 0.4/1+0.6/2+0.7/3+0.8/4 and B = 0.3/1+0.65/2+0.4/3+0.1/4 Calculate the fuzzy union $A \cup B$, the fuzzy intersection $A \cap B$, and the complement of set A.	5
2	What is meant by "defuzzification," and what are the most widely used techniques for defuzzifying in fuzzy logic systems?	5
3	Mention the importance of objective (fitness) function in Genetic Algorithm (GA).	5
4	Discuss the pros and cons of Ant Colony Optimization (ACO).	5
5	What are the key components of Particle Swarm Optimization (PSO) that determine the algorithm's performance?	5
	PART B (Answer any five questions 5*7=35 marks)	
6	Let V = {A, B, C, D} be the set of four kinds of vitamins, F = {f_1, f_2, f_3} be three kinds of fruits containing the said vitamins to various extents, and D = {d_1, d_2, d_3} be the set of three diseases that are caused by deficiency of the vitamins. Vitamin contents of the fruits are expressed with the help of the fuzzy relation R over F × V, and the extent to which the diseases are caused by the deficiency of these vitamins is given by the fuzzy relation S over V × D. Relations R and S are given below. $R = \begin{array}{c c} & \begin{matrix} A & B & C & D \end{matrix} \\ \begin{matrix} f_1 \\ f_2 \\ f_3 \end{matrix} & \begin{bmatrix} 0.5 & 0.2 & 0.1 & 0.1 \\ 0.2 & 0.7 & 0.4 & 0.3 \\ 0.4 & 0.4 & 0.8 & 0.1 \end{bmatrix} \end{array}$, $S = \begin{array}{c c} & \begin{matrix} d_1 & d_2 & d_3 \end{matrix} \\ \begin{matrix} A \\ B \\ C \\ D \end{matrix} & \begin{bmatrix} 0.3 & 0.5 & 0.7 \\ 0.1 & 0.8 & 0.4 \\ 0.9 & 0.1 & 0.2 \\ 0.5 & 0.5 & 0.3 \end{bmatrix} \end{array}$ Assuming that the max–min composition $R \circ S$ represents, in some way, the correlation between the amount of a certain fruit that should be taken while suffering from a disease, find $R \circ S$.	7

7	Using the inference approach, find the membership values for the fuzzy triangular shapes (i) isosceles triangle (ii) equilateral triangle (iii) right angle triangle (iv) isosceles and right angle triangle.	7
8	Discuss on Mamdani fuzzy inference system with a suitable example.	7
9	Consider a simple Genetic Algorithm (GA) used to solve an optimization problem where each individual in the population represents a candidate solution. Assume the chromosome is a 6-bit binary string and the goal is to maximize the following objective function: $f(x)=2x_1+3x_2+4x_3+5x_4+6x_5+7x_6$ where x_1, x_2, x_3, x_4, x_5 and x_6 are the genes of the chromosome. The initial population of the chromosomes is shown below, along with their fitness values: (see table below) If one point crossover method is used, derive two possible populations for the next generation. Assume that the crossover and mutation probabilities are 0.8 and 0.05, respectively.	7

Chromosome ID	Chromosome (Binary)	Fitness Value
1	110101	27
2	101110	26
3	011011	18
4	111100	30
5	010101	20

10	Describe the different phases in the life cycle of a genetic algorithm, and how do processes like encoding, selection, crossover, mutation, and evaluation work together to evolve solutions through successive generations.	7
11	Consider a Traveling Salesman Problem (TSP) where you need to visit 5 cities, starting and ending at city A. The distance matrix between the cities is as follows: (see table below) The goal is to find the shortest path that visits each city exactly once and returns to the starting city. Solve TSP using Ant Colony Optimization (ACO). Given the evaporation factor ρ =0.02 and initial pheromone at all edges τij=100.	7

City	A	B	C	D	E
A	0	10	15	20	25
B	10	0	35	25	30
C	15	35	0	30	5
D	20	25	30	0	15
E	25	30	5	15	0

12	Consider an optimization problem whose objective function is given as $f(x) = x^2 - 6x + 8$. Minimize this function over the search space $x \in [-10,10]$, using Particle Swarm Optimization (PSO). There are 4 particles in the swarm. The initial positions of the particles are: $x_1 = -7$, $x_2 = 2$, $x_3 = 5$, $x_4 = -3$ and initial velocity, $v_i = \{0\}$. Assume that both the random numbers r_1 and r_2 used for computing the movement of the particle towards the individual best position and social best position are 0.5 and the accelerating coefficients, c_1 and c_2 are 0.2. What would be the next position of each particle after two iterations of the PSO algorithm if the inertia parameter ω that is used along with current velocity update formula is 0.8?	7